

BAIE-COMEAU, QC, Canada

WMO: 718290

Lat: 49.259N

Lon: 68.146W

Elev: 425

StdP: 14.47

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.3	-10.0	-26.7	1.3	-12.7	-21.9	1.7	-8.7	34.1	9.4	29.9	9.0	13.3	310	0.705

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	77.2	63.3	74.1	61.9	71.1	60.4	66.5	73.3	64.4	70.4	62.6	68.0	12.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	90.1	69.5	62.0	84.5	67.1	60.3	79.4	65.1	31.4	73.5	29.8	70.4	28.5	68.0	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.5	23.1	20.4	DB	-19.8	83.5	3.8	2.5	-22.6	85.3	-24.8	86.7	-26.9	88.1	-29.7	90.0	
			WB	-20.4	70.3	3.6	2.2	-23.1	71.9	-25.2	73.2	-27.2	74.4	-29.9	76.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	9.5	12.4	22.1	34.7	46.7	56.5	62.4	61.4	53.4	41.9	30.8	17.6
	DBStd	19.93	11.49	11.06	10.44	6.89	6.38	6.14	4.45	4.53	6.07	6.15	7.78	10.26
	HDD50	5659	1254	1052	865	459	144	11	0	0	31	261	578	1004
	HDD65	10058	1719	1472	1330	909	568	265	106	128	350	715	1027	1469
	CDD50	1131	0	0	0	0	42	205	384	355	134	11	0	0
	CDD65	57	0	0	0	0	1	9	26	18	3	0	0	0
	CDH74	270	0	0	0	0	9	67	109	72	13	0	0	0
	CDH80	23	0	0	0	0	1	10	9	3	0	0	0	0
Wind	WSAvg	10.0	12.0	12.0	11.5	10.7	9.0	8.5	7.8	8.0	8.7	10.0	10.8	11.5
Precipitation	PrecAvg	39.8	2.4	2.2	2.7	3.1	3.1	3.7	3.5	3.6	3.7	3.8	3.7	3.7
	PrecMax	49.9	4.2	5.6	4.6	5.5	5.0	6.1	4.9	9.4	9.1	7.5	5.8	7.1
	PrecMin	33.3	0.6	1.2	1.3	0.7	1.3	2.1	0.9	1.1	0.9	1.7	1.1	2.0
	PrecStd	4.5	1.1	1.1	0.9	1.5	1.1	1.1	1.3	2.0	1.8	1.5	1.2	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.0	39.5	48.6	57.7	74.7	80.9	80.6	79.8	75.8	64.1	52.1	41.7
		MCWB	38.0	37.5	38.6	46.1	56.1	63.8	66.3	66.6	64.2	54.0	49.4	40.1
	2%	DB	35.1	35.8	41.5	51.1	66.9	75.1	77.1	75.9	70.2	58.4	47.2	37.9
		MCWB	33.5	33.4	35.9	41.8	52.8	60.4	63.8	63.5	60.9	52.6	44.1	36.2
	5%	DB	31.1	31.8	38.3	47.3	61.1	70.8	73.9	72.7	66.3	54.3	43.8	34.9
		MCWB	29.7	29.6	33.9	39.7	49.6	58.1	62.4	62.1	58.6	49.1	40.7	33.3
	10%	DB	26.5	28.6	35.6	44.1	56.8	66.9	70.9	69.6	62.9	51.0	41.0	32.1
		MCWB	25.0	26.6	32.0	37.7	47.9	56.3	61.0	60.0	56.5	46.3	38.4	30.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.3	38.2	41.3	48.1	59.6	66.8	69.8	68.6	66.5	57.4	50.3	40.8
		MCDB	38.7	39.0	44.1	54.3	68.3	75.0	76.4	76.9	72.8	60.9	52.1	41.4
	2%	WB	33.7	33.7	37.1	43.5	55.1	63.1	67.0	65.7	62.7	53.6	45.0	36.7
		MCDB	34.8	35.1	40.3	48.8	63.1	70.3	73.5	72.7	67.9	57.5	46.3	37.5
	5%	WB	29.6	29.9	34.7	40.8	51.9	60.6	64.7	63.7	59.8	50.0	41.2	33.5
		MCDB	30.9	31.8	37.5	46.0	58.5	66.9	70.3	69.5	64.7	53.0	42.9	34.7
	10%	WB	25.1	26.8	32.2	38.5	49.3	58.3	62.8	62.1	57.4	46.9	38.7	30.8
		MCDB	26.3	28.4	35.5	43.0	55.2	64.9	68.2	67.0	61.8	50.3	40.7	32.1
Mean Daily Temperature Range	5% DB	MDBR	14.9	15.3	14.9	12.9	14.2	14.3	13.9	13.6	13.4	10.6	9.5	12.3
		MCDBR	15.3	13.4	14.9	17.4	22.0	21.1	19.0	18.5	17.8	15.2	11.4	12.2
		MCWBR	15.3	12.1	11.4	11.3	11.9	10.7	9.5	9.5	11.3	10.7	10.0	11.8
	5% WB	MCDBR	15.7	12.8	14.1	15.4	18.1	17.4	16.1	15.4	15.9	14.2	10.8	12.2
		MCWBR	15.9	12.0	11.9	11.3	11.9	10.7	9.9	9.4	11.6	11.6	10.5	12.1
Clear-Sky Solar Irradiance	taub	0.263	0.269	0.299	0.352	0.366	0.391	0.417	0.395	0.371	0.328	0.310	0.266	
	taud	2.333	2.312	2.283	2.270	2.362	2.357	2.316	2.376	2.421	2.522	2.401	2.381	
	Ebn at Noon	255	282	290	283	283	276	266	267	262	258	233	238	
	Edh at Noon	21	28	34	39	37	37	39	35	30	23	20	18	
All-Sky Solar Radiation	RadAvg	386	698	1096	1436	1653	1750	1671	1429	1081	637	343	259	
	RadStd	43	69	83	120	135	132	164	85	78	55	35	24	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (14 neighbors)	N/A	N/A	+2.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page